

Task1: Java Basics & OOP Fundamentals

Code:

```
import java.util.Scanner;

public class calculator {

    public static void main(String[] args) {

        try (Scanner sc = new Scanner(System.in)) {

            System.out.print("Enter first number: ");

            double num1 = sc.nextDouble();

            System.out.print("Enter second number: ");

            double num2 = sc.nextDouble();

            System.out.println("\nCalculator Menu");

            System.out.println("1. Addition");

            System.out.println("2. Subtraction");

            System.out.println("3. Multiplication");

            System.out.println("4. Division");

            System.out.print("Enter your choice: ");

            int choice = sc.nextInt();

            switch (choice) {

                case 1:

                    System.out.println("Result = " + (num1 + num2));

                    break;

                case 2:

                    System.out.println("Result = " + (num1 - num2));

                    break;

                case 3:

                    System.out.println("Result = " + (num1 * num2));
```

```
break;

case 4:

if (num2 != 0) {

System.out.println("Result = " + (num1 / num2));

} else {

System.out.println("Division by zero is not allowed.");

}

break;

default:

System.out.println("Invalid Choice!");

}

sc.close();

}

}

}
```

OOPS Concepts Used:

- Scanner Class (User Input)
- Variables and Data Types
- Switch Statement
- Arithmetic Operators (+, -, *, /)
- If-Else Condition
- Output Statements (System.out.println())

Output:

<https://github.com/isha05-alt/calculator>

```
Enter first number: 3
Enter second number: 1
```

```
Calculator Menu
1. Addition
2. Subtraction
3. Multiplication
4. Division
Enter your choice: 4
Result = 3.0
```